Magnetic fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Interpret magnetic field patterns and compass direction.

SUCCESS CRITERIA - I can...

1. Label north and south poles.
2. Trace external field direction from north toward south.
3. Explain how field-line drawings represent a field and what they leave out.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “magnetic field” and “pole” to explain your answer. Sentence starters are optional.

magnetic field: a field that influences magnets and moving charges

pole: a north or south region of a magnet

superposition: adding displacements or fields at the same place and time

01 Warm-up: bring back an earlier idea

A total current of **10 A** splits between two identical parallel branches. **Find each branch current**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $B_{net,\,x} = B_{1,\,x} + B_{2,\,x}$; $B_{net,\,y} = B_{1,\,y} + B_{2,\,y}$.
Add field components at the same point. A compass follows the net field. Outside a bar magnet, arrows run *north* to *south*.

Worked reasoning: Outside a bar magnet, field arrows run from the *north* pole toward the *south* pole. A compass's *north*-seeking end follows the local field direction.

Your turn: Draw curved external field lines and two small compass arrows at different points.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Label N** and S on the bar magnet.
2. **Sketch external field arrows** and the direction a nearby compass would point.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

04 Apply the physics

Place two bar magnets on a table without letting them touch.
1. **Predict the interaction** when N faces N, then when N faces S.
2. **Sketch both arrangements** and **draw the force** on each magnet.

Sentence starter: I predict ... . I used ... to make this prediction because ... .

Label poles; add force or field arrows.



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student draws field lines that cross at one point. **Explain why this** is inconsistent with a single net field direction at that point.

Sentence starter: I think … . I can point to …, which shows …, so … .

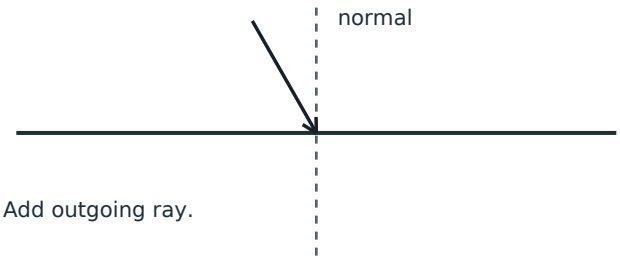
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A ray enters a medium with speed **1/4** of its original speed at an angle. **Sketch the bend** and state whether frequency changes.
Label both media, the normal and ray directions.

Word help: frequency = number of cycles each second

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

07 Exit check

What does closer spacing of field lines conventionally represent?

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Cut a bar magnet into two pieces in a drawing. **Predict the poles** on each piece and **show the external field pattern** qualitatively.

Sentence starter: I predict … . I used … to make this prediction because … .



Marzano check - circle: ① ② ③